

Phase 6 - Booking & Payments

Time	Start date	Finish date	Status
2 weeks			Not started

Goal: Students can book and pay safely - with no overselling, ever.

Real-life example: IRCTC tatkal booking - thousands of people click at the same moment, but a seat must never be given to two people.

Step checklist

#	Step	Done
1	Flyway V3: bookings and payments tables	☐
2	Book API creates a HELD booking with a 10-minute timer and reduces seats	☐
3	Optimistic locking with @Version on Event; retry when two requests clash	☐
4	Scheduled job every minute: expire old holds and give the seats back	☐
5	Free events confirm immediately	☐
6	Stripe Checkout session for paid events (test mode)	☐
7	Stripe webhook with signature check confirms the booking (Stripe CLI for local testing)	☐
8	Idempotency - the same webhook twice never confirms twice	☐
9	Cancel booking API; seats come back	☐
10	Concurrency test: 100 parallel requests on 10 seats = exactly 10 bookings	☐
11	React: Book button, countdown timer, Stripe redirect, success / cancel pages, My Tickets with status	☐

Output of this phase

- Working booking engine with seat hold
- Stripe test payments
- Passing concurrency test

Done when (check before moving on)

Check	Passed
Test card 4242 4242 4242 4242 confirms a booking	☐
An unpaid hold expires and the seat returns	☐
The concurrency test passes every time	☐

Skills you practise

@Transactional, optimistic locking, @Scheduled, webhooks, idempotency, multithreading

Where your work goes

backend\...\booking\ and payment\; frontend checkout and My Tickets pages

My notes
